

Lab 5: Packages and Built-in Datasets

Level: Intermediate Mapping: LO3 | K3 | A2 Concepts: packages, namespaces, documentation, str, summary

Scenario

A junior analyst must document a reproducible package and dataset setup.

Goal

A setup script and mtcars/quakes data profile.

Files in this folder

- lab-05.R - completed guided R script
- verify.R - repeatable acceptance check
- lab-workbook.xlsx - brief, input data, controls and evidence record
- input-data.csv - portable fallback if readxl is unavailable

Detailed step-by-step

1. Review package requirements in the workbook.
2. Check for readxl without reinstalling it on every run.
3. Load and document mtcars and quakes.
4. Create a compact data profile with rows, columns, missingness and ranges.
5. Write the profile to CSV and log package versions.

Run and verify

1. Set the RStudio working directory to this lab folder.
2. Open lab-05.R and read the header, inputs and validation rules.
3. Run the complete script with `source("lab-05.R")`.
4. Confirm the Console ends with LAB VERIFIED: outputs created and checks passed.
5. Run `source("verify.R")` and retain the verification message.
6. Open analysis_output.csv and compare the evidence with the workbook Control sheet.
7. Complete the workbook Evidence sheet and save it without changing the filename.

Acceptance criteria

- [] Script uses `requireNamespace()` before optional installation.
- [] Data profile reports correct dimensions.
- [] Package and R versions are recorded.

Troubleshooting

- there is no package called 'readxl': the script automatically uses input-data.csv; install readxl later if Excel import is required.

- cannot open file: confirm RStudio's working directory is this exact lab folder.
- object not found: run the script from the top so imports and validations occur before analysis.
- Verification fails: inspect the first error, compare column names with Input_Data, and rerun from a clean session.

Evidence to submit

Completed Excel Evidence sheet, R script, generated output files and the final Console verification message.